



EAST WEST UNIVERSITY
Department of Mathematical & Physical Sciences
Semester: Summer – 2025

Course Code: MAT 104 Title: Co-ordinate Geometry and Vector Analysis

Course Instructor: Dr. Md. Shafiqul Islam, Professor of Applied Mathematics, DU
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Course Outline – Section 06

Class Hours: TR 03:10 – 04:40 (Room: FUB 603)

Consultation: TR: 01:30 – 03:00 pm (and by appointment), Room: FUB 1004

Mathematics, as one of the great triumphs of the human intellect, brings us together as human beings interested in partaking of the best of the human spirit. The study of geometry using coordinate points is known as coordinate geometry (or analytic geometry). It is a portion of geometry in which the position of a point is determined by coordinates. A series of values called coordinates is used to show the exact location of a point in the coordinate plane. It's always better to understand how information benefits us in the real world. Let's look at how the coordinate geometry can be applied in practice. A few examples of coordinate geometry **applications and outcomes** are:

- The text file or PDF file is an example of coordinate plane usage. The words or graphics in these files are written or edited using coordinate geometry. A 2-dimensional coordinate (x, y) system is used to read any PDF file that comprises text and graphics.
- Coordinate geometry is also used by scanners and photocopying machines to generate a perfect copy of the original image. It's used in physics, GPS, maps, and a variety of other fields under various names.
- Even a computer monitor employs some basic coordinate geometry concepts. Complex curves, conics, and shapes can be mathematically specified using algebraic equations, which would be impossible to analyze using pure geometry.
- Coordinate geometry is used to manage and regulate air traffic. The coordinates of the flight are used to describe the aircraft's current location. Even if an aircraft moves a tiny distance (up, down, forward, or backward), the system updates the coordinates of flight for every small change in its position.
- Vectors are used for EVERYTHING. Vectors are the heart and soul of Cartesian geometry and therefore the bulk of physics (at least mechanics, Fluid mechanics, thermodynamics, control/signals, etc).
- Computer graphics is based on rendering, which calculates the Red/Green/Blue vector components of the pixels on your screen. If you are using a screen to read this, the 'framebuffer' is a vector that holds the value of all those pixels you are looking at. Every digital picture you look at is a vector of pixels. Digital music uses vectors to mix different sounds together and calculate synthetic sounds. So, whether or not you use vectors in your job, you almost certainly are affected by vectors in the systems we depend on every day.

To be a successful engineer or scientist, you will be required to solve problems in your job that you have never seen before. It is important to learn problem-solving techniques, so that you may apply those techniques to new problems. A common mistake is to expect to learn some prescription for solving all the problems you will encounter in your later career.

However, this course will help to develop the student's analytical and creative ability for solving different engineering problems, such as electrical networks, frameworks in mechanics, curve fitting and other optimization problems, systems of differential equations, partial differential equations, fluid dynamics, quantum theory, etc. Thus, the objectives of this course are:

- ✓ Engage students in sound mathematical thinking and reasoning. This should include students finding patterns, generalizing, and asking/answering relevant questions.

- ✓ Learning how to formulate the mathematical model and how to interpret the results is what your physics and engineering classes do. In this course, we will focus mostly on the mathematical analysis. Sometimes we will work with simple real-world examples so that we have some intuition and motivation about what we are doing.
- ✓ Provide a setting that prepares students to read and learn mathematics on their own.
- ✓ Explore multiple representations of topics including graphical, symbolic, numerical, oral, and written. Encourage students to make connections among the various representations to gain a richer, more flexible understanding of each concept.
- ✓ Analyze the structure of real-world problems and plan solution strategies. Solve the problems using appropriate tools.
- ✓ Develop a mathematical vocabulary by expressing mathematical ideas orally and in writing.
- ✓ Enhance and reinforce the student's understanding of concepts through the use of technology when appropriate.

Contents of MAT 104

SI	Topics	Remarks
1.	Two-Dimensional Geometry: Transformation of coordinates and axes. Pair of straight lines	Mid
2.	Circles: Tangent and Normal, Chord of contact, System of circles, Orthogonal circles.	
3.	Conic Section: The general equation of second degree, Parabola, Ellipse, and Hyperbola	
4.	Three-Dimensional Geometry: Co-ordinate system. Direction cosines and direction ratios.	
5.	Plane, Straight line. The shortest distance.	Final
6.	Sphere: Tangent plane. Cylinder and Cone	
7.	Vector Analysis: Algebra of vectors, Vector differentiation and vector integration.	
8.	Gradient, Divergence, and Curl. Physical significance of Gradient, Divergence, and Curl.	
9.	Green's Theorem, Divergence Theorem, and Stoke's Theorem	
10.	Cartesian, Spherical, Polar, and Cylindrical system	

References:

1. University Analytic and Vector Geometry (2D & 3D, Coordinates and Vectors): **Md. Abdur Rahman**
2. Vector Analysis: Murray R Spiegel (**Schaum's Outline Series**)
3. A text Book of Two-Dimensional Geometry: Sat Pal and Harbans Lal
4. Calculus: Howard Anton (10th Edition, Chapter 15, Topics in Vector Calculus)

Exam Schedule and Marks Distribution (Total 100)

Exam Schedule and Marks				Class Assessments	Marks
Examination	Day	Date	Marks	Class Attendance	10
Mid Term	Thursday	22 July (Probable)	30	Assignments/Home works	10
Final	Tuesday	09 Sep 2025	30	Quizzes (At least two)	20
Subtotal			60	Subtotal	40

Thank you.